Would It Replicate?

Team Members: _____ _____ _____

_____ _____

Everything in this activity is simulated. The creek, its sampling sites and every chloride value come from a random number generator. It is set up like your Pennypack upstream/downstream chloride comparison, but it is **not** the Pennypack data and says nothing about your Pennypack result.

Part 1 — The Report

> **Team A:** We measured chloride at 12 randomly chosen sites upstream of a stormwater outfall and 12 randomly chosen sites downstream. Downstream sites averaged **35 mg/L higher**. A test of the difference in means gives **p = 0.03**.

Team B will now repeat the survey exactly: same creek, same season, 12 sites on each side, with a new random choice of sites.

1. **What is the chance that Team B's survey also comes out with p < 0.05? Each team member writes a number on their own, then compare.**

   Me: _____ % Teammates: _____ % _____ % _____ %

2. **What did you use to get your number?**

   _____

3. **Suppose a downstream excess is real. What is the smallest excess that would be worth acting on, for example worth asking the township to look at the outfall? Commit to a number now.**

   _____ mg/L because _____

STOP. Every team member answers 1–3 before anyone opens the notebook.

Part 2 — Run the Creek

In the notebook you can do something no field team can: survey the same creek as many times as you like. You also know the creek's truth. Downstream sites average **33 mg/L** more chloride than upstream sites, and individual sites scatter around their average with an SD of 40 mg/L.

4. ***After 2.4 at 10, before you change it to 1000:*** **Predict what fraction of 1000 surveys of this creek will give p < 0.05.**

   Prediction: _____ *After 2.5:* actual _____ smallest p _____ largest p _____

   The creek never changed. What did? _____

Part 3 — Only the Surveys Like Team A's

The notebook now runs the creek in pairs: an *original* survey and a *replicate* of it. Then it keeps only the pairs whose original came out like Team A's, with p between 0.01 and 0.05.

5. ***Predict, before 3.2:*** **Among the replicates of those kept pairs, the fraction with p < 0.05 will be (circle one) compared with your answer to question 4:**

   HIGHER ABOUT THE SAME LOWER because _____

6. ***After 3.2:***

   Pairs kept: _____ of 1000 Their replicates with p < 0.05: _____

   All 1000 replicates with p < 0.05: _____

7. ***After 3.3:*** **A survey gets p < 0.05 when its observed difference is more than _____ mg/L from zero. The creek's true difference is 33 mg/L. Use those two numbers to explain (a) the fraction you found in question 4, and (b) why an original p = 0.03 does not change the replicate's chances.**

   (a) _____

   _____

   (b) _____

   _____

Part 4 — More Sites

Same creek, same 33 mg/L excess, same SD. Only the number of sites changes.

Our team's sites per side: _____

8. ***Before 4.3 at 1000:*** **predict the fraction of surveys with p < 0.05, then record what you got.**

	Prediction	**Actual**
Real creek (excess 33 mg/L)		
No-excess creek (0 mg/L)		

Cutoff from 4.2: a survey needs a difference farther than _____ mg/L from zero.

Report your two actual fractions to the board, then copy the whole class's results here.

Sites per side	**6**	**12**	**24**	**48**
Cutoff (mg/L)				
Real creek: fraction p < 0.05				
No-excess creek: fraction p < 0.05				

9. **More sites changed one row of fractions a lot and the other hardly at all. Which is which? Explain using the cutoff row.**

Part 5 — How Many Discoveries Are False?

Imagine 1000 creeks with outfalls, each surveyed once. In 100 of them the excess is real (33 mg/L). In the other 900 there is no excess at all. Nobody knows which creeks are which; they only see each survey's p-value.

10. Fill in the table using the class results from Part 4. Round to whole creeks.

	6 sites per side	**48 sites per side**
Real creeks with $p < 0.05$ (100 × real-creek fraction)		
No-excess creeks with $p < 0.05$ (900 × no-excess fraction)		
Total creeks with $p < 0.05$		
Share of those with **no** real excess		

11. A classmate says: "p < 0.05 means there's at most a 5% chance the effect isn't real." Use your table to answer them. What besides the number of sites would change the bottom row?

Part 6 — Twenty Surveys, Twenty Intervals

The prebuilt cell runs twenty surveys of the 12-site creek. Each one gets a 95% bootstrap confidence interval (Class 19) and a p-value. Every team sees the same twenty surveys. Answer from the plot and table only; you do not need to read the code.

12. Count, out of 20:

Intervals containing 33: _____ $p < 0.05$: _____ intervals entirely above 0: _____

One survey's interval and p-value disagree about whether there is a difference. Which? _____

For that survey, compare its observed difference with the cutoff you wrote in question 7. The p-value judged the survey against surveys of a creek with *no* excess, built with the creek's true SD. The interval used only that survey's own 24 sites. Explain how the two can disagree.

Part 7 — A Very Big Survey

Look at the screen. A different creek has a conductivity logger on each side of its outfall: **20,000 readings upstream and 20,000 downstream**. The SD is still 40 mg/L.

13. Answer each part *before* the next reveal.

(a) Histograms: is downstream different? YES NO CAN'T TELL

(b) The p-value will be: LARGE NEAR 0.05 TINY

(c) 95% interval: _____ to _____ mg/L.

Compare with your answer to question 3. Is this excess worth acting on? Why?

_____

_____

14. Back to Part 1. In two or three sentences, tell Team B what Team A's p = 0.03 does and does not tell them, about their own survey and about how big the excess is.

_____

_____

_____

If You Finish Early

A. The average observed difference among the *significant* surveys was _____ mg/L. The truth is 33. Suppose a journal only prints surveys with p < 0.05. What happens to the differences readers see in print?